Robust Image Classification in the Wild

The Journey from Simple CNNs to a Self-Refining Ensemble

Group 24

EE708

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Core Ideas: BroadResNet, Data Mixing, & Pseudo-Label Propagation

The Challenge: 32x32 Pixels & Uncharted Territory

Dataset Constraints

- **Low Resolution:** 32×32 pixel images. The network has highly restricted spatial information to extract features from.
- **Severe Class Imbalance:** Unequal representation across the 10 categories, heavily skewing standard learning trajectories.
- **Distribution Shift:** The test set introduces unknown corruptions and variations absent from the clean training data.

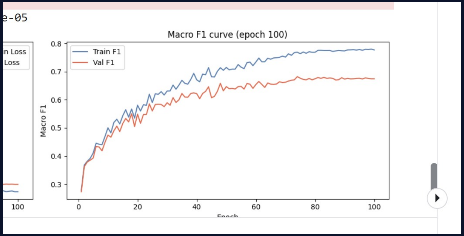
The Starting Line

- **First Foray:** Tackling a robust image classification challenge of this scale required an iterative approach.
- **The Goal:** Before building complex ensembles, we needed to verify if we could extract meaningful features from skewed, low-resolution data.
- **The Strategy:** Establish a simple baseline, test its capacity to overfit, and progressively apply regularization.

Iteration 1: The Basic CNN & The Generalization Gap

Baseline: 3-Layer CNN

- **Architecture:** A lightweight, custom 3-layer convolutional network.
- **Complexity:** Highly constrained at **289,194 parameters**.
- **Training Setup:** 100 epochs across 184 training batches
- **The Result:** The network achieved a Train F1 of ≈ 0.80 that is failed to even overfit
- **Observation:** The model lacked the structural capacity to handle features in the data.

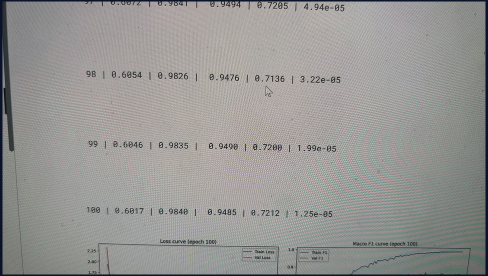


Top: Training validation logs for the SimpleCNN baseline.

Iteration 2: Scaling the Custom Baseline

Scaling Depth & Spatial Logic

- **Spatial Tuning:** Increased model capacity by adding CNN layers. Optimized kernel sizes, strides, and padding to capture finer feature hierarchies.
- **Capacity Jump:** Scaled up to **1,152,810 parameters**.
- **Training Peak:** Achieved a near-perfect **0.98 Train Macro F1**.
- **The Barrier:** Validation F1 plateaued at **0.72**, indicating severe overfitting and necessitating rigorous regularization.

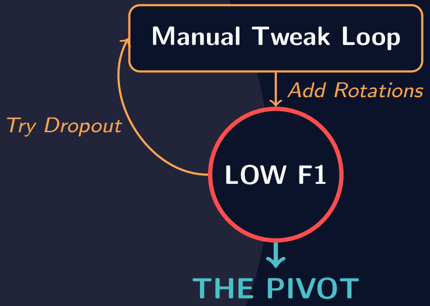


Evidence of Overfitting

Phase 3: The Augmentation Paradox

Fighting the "Brittle" Nature

- **Manual Regularization Stack:**
  - Aggressive Dropout ($p = 0.7$).
  - Weight Decay (L_2 penalties).
- **Aggressive Augmentation:** Applied Rotations, Flips, and Random Crops.
- **The Paradox:** Instead of generalizing, the model's test performance plummeted.
- **Diagnosis:** The augmentations were counterproductive. Without sufficient parameter capacity, the CNN could not distinguish between useful invariant features and augmentation noise.



"We weren't training a model; we were confusing it."

Phase 4: Entering the Wide Regime

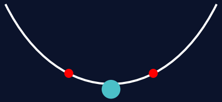
The BroadResNet-28-10

- **Shallow Depth (28 Layers):** Since our input images are small (32×32), going deeper than 28 layers with pooling would shrink the spatial dimensions too much, essentially collapsing the image before it can learn useful patterns.
- **Widening Factor ($k = 10$):** To make up for the shallow depth, we multiply the number of filters by 10. This allows the network to extract a massive amount of features directly from the 32×32 resolution, resulting in 36.5 million parameters.
- **Frequency Adjusted Loss:** This loss function adjusts the penalty based on how common a class is. It ensures the model pays extra attention and learns better on the less frequent images, rather than just optimizing for the majority classes.

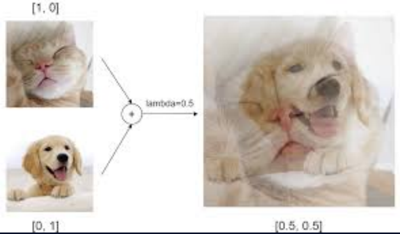
Phase 5: Smoothing the Decision Boundary

Beyond Standard Training

- **Stochastic Weight Averaging (SWA):** Instead of saving just the final model, we average the weights across the last 60 epochs. **Why?** It prevents the model from getting stuck in sharp, fragile performance valleys, finding a flatter minimum that handles unseen test data better.
- **Data Mixing (Mixup & PatchMix):** Training on artificially blended images—fading two images together or copy-pasting patches. **Why?** It forces the network to learn broad, robust features instead of memorizing specific pixels, making it resilient to corrupted inputs.



SWA Optimum (Flatter)



Visualizing the Mixup Strategy

Phase 6: Multi-View Reasoning (The Ensemble)

Reducing Variance

- **2-Seed Ensemble:** Trained two entirely independent BroadResNets (Seeds 42 & 137) to capture diverse weight initializations.
- **Test-Time Augmentation (TTA):** Each test image was evaluated **13 times** rather than just once.
- **Strategy:** Computed predictions across 12 random augmentations (Crops, Flips, Color Jitter) plus the original image.
- **Outcome:** Averaging these views filtered out anomalous ”hallucinations,” yielding a highly stable, calibrated probability distribution.

TTA Views $\times 13$

$\boxed{\text{Test Image}} \longrightarrow \boxed{\sum \text{Ensemble}}$

Confidence Filtering:
Strict Threshold $= 0.90$
for Pseudo-Labeling Phase

The Final Push: Self-Refining Pseudo-Labels

Adapting to the Test Distribution

- **The Loop:** Extracted ultra-high confidence test predictions ($> 90\%$ confidence) from the ensemble.
- **Label Propagation:** Appended these high-confidence samples back into the training set as pseudo ground-truth labels.
- **Fine-Tuning:** Re-trained the model for an additional 15 epochs on this refined, combined dataset.
- **Why it Worked:** This self-training mechanism effectively bridged the domain gap, allowing the model to adapt directly

Mission Accomplished

- **The Lesson:** Scaling capacity is necessary, but architectural design and regularization (SWA, Mixup) are the true differentiators.
- **The Final Pipeline:** BroadResNet-28-10 + SWA + Data Mixing + Pseudo-Labeling.
- **Final Standing:** Achieved a highly competitive **0.93 Macro F1** on the Public Leaderboard.

Q & A — Group 24